Practice Set (2026)
Discrete Mathematics (PCC-CSE 404)

Module-I

1. Let ρ e a relation on the set C of all complex numbers and is defined by $(a_1 + ib_1)\rho(a_2 + ib_2)$ if and only if $a_1 \leq a_2$ and $b_1 \leq b_2$ for all $a_1 + ib_1, a_2 + ib_2 \in C$. Prove that (C, ρ) is a POSET.
2. Let $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,1), (1,3), (3,1), (2,3), (3,2)\}$ be a relation defined on $A = \{1,2,3,4\}$. Prove that R is an equivalence relation and hence find the equivalence classes.
3. A relation is defined by $\rho = \{(a, b): a, b \in Z$ and $|a - b| \leq 5\}$. Show that ρ is a equivalence relation.
4. If $f, g: \mathbb{R} \to \mathbb{R}$ be two functins definde by $f(x) = x^2 + 3x + 1$ and $g(x) = 2x - 3$ then find $(f \circ g)(x)$ and $(g \circ f)(x)$
5. Show that the functions $f: \mathbb{R} \to \mathbb{R}, f(x) = x^3$ is a bijective. Also find the inverse function of f.
6.
7. Find the domain of definition of the function $\sqrt{\frac{1-|x|}{2-|x|}}$.
8. Find the range of the function $f(x) = \log_2\left(\frac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}}\right)$.
9. Find the range set of the function $f(x) = \frac{x}{x^2+1}$.
10. Draw the Hasse diagram of the Poset $(A, /)$where $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$.
11. Draw the Hasse diagram of the Poset (S_{30}, R) where S_{30} is the set of all positive divisors of 30 and R is the relation division.
12. Draw the Hasse diagram of the Poset (D_{40}, ρ) where D_{40} be the set of all positive integers of 40 and ρ is the relation division.
13. Draw the Hasse diagram of the Poset $(S, \ /)$ where $S = \{2, 3, 5, 30, 60, 120, 180, 360\}$.
14. Draw the Hasse diagram of the Poset $(A,\ R)$ where $A = \{2, 3, 5, 10, 12, 15, 18, 20, 30, 45, 60, 75, 90\}$ and R is the relation division.
15. If $gcd(a, b) = 1$, prove that $gcd(2a + b, a + 2b) = 1\ or\ 3$.
16. If $gcd(a, 4) = 2$ and $gcd(b, 4) = 2$ then prove that $gcd(a + b, 4) = 4$.
17. If $gcd(a, b) = 1$ then prove that $gcd(a + b, a - b) = 1\ or\ 2$.
18. Show that the square of any odd integers is of the form $8k + 1, k \in Z^+$.
19. Show that every square integer is of the form $9k\ or\ 3k + 1, k \in Z^+$.
20. Show that the cube of any integer is of the form $9k, 9k \pm 1, k \in Z$.
21. Show that the product of any three consecutive integers is divisible by 3!.
22. Show that the product of any r consecutive natural numbers is divisible by $r!$.
23. Find all the solutions in positive integers of $5x + 3y = 52$.
24. Solve the diophantine equation $158x - 47y = 9$.
25. Find all solution in positive integers: $12x + 201y = 1$.
26. Find the solutions of the equation $243x + 198y = 9$.
27. Show that $3^{302} \equiv 4\ (mod\ 5)$.

28. Prove that $11^{34} \equiv 2\ (mod\ 17)$.
29. Find the reminder when 7^{32} is divided by 5.
30. Find the unit digit in 3^{909}.
31. Find the unit digit in 13^{37}.
32. Show that $2^{18} + 9^{18} \equiv 0\ (mod\ 17)$.
33. Solve the linear congruence $6x \equiv 3(mod\ 9)$.
34. Solve the linear congruence $9x \equiv 21(mod\ 30)$.
35. Solve the linear congruence $18x \equiv 30(mod\ 42)$.
36. Solve the linear congruence $345x \equiv 18(mod\ 912)$.
37. Find the inverse of 19 modulo 141.
38. Find the inverse of 144 modulo 233.
39. Find all integers $m \geq 5$ such that $7 \equiv m^2\ (mod\ m)$.
40. Prove that if an integer n is the sum of two squares $(n = a^2 + b^2\ \forall a, b \in Z)$ then $n = 4q$ or $4q + 1$ or $4q + 2$ for some $q \in Z$. Hence deduce that 1234567 can not be written as sum of two squares.

Module-2

1. Find the number of words that can be formed from the letters of the word "ELEVEN" (a) How many of them begin and end with E? (b) How many of them have the three E's together? (c) How many begin with E and end with N?
2. How many ways are there for 8 men and 5 women to stand in a line so that no two women stand next to each other?
3. Find the number of ways in which 4 IT students and 17 CSE students can sit together in a queue such that between any two IT students at least two CSE students can sit.
4. What is the rank of the word "GOOGLE" written down in a dictionary.
5. What is the rank of the word "MODESTY" written down as in a dictionary.
6. Find the number of combinations of 10 items taking from five x's, four y's, five z's and seven w's.
7. Find the number of ways a company can assign 6 projects to 4 employees so that each employee gets at least one project
8. Find the number of divisors of the number 2160. Also, find the sum of these divisors.
9. How many solutions of the equation $x_1 + x_2 + x_3 = 11$ have, where x_1, x_2, x_3 are non-negative such that $x_1 \leq 3, x_2 \leq 4$ and $x_3 \leq 6$?
10. In how many ways 10 students can be divided into three teams, one containing four students and the other two contains three each.
11. State Generalizes Pigeonhole Principle? 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, what is the minimum number of different rooms that will be needed?
12. Find the minimum number n of integers to be selected from $S = \{1, 2, \ldots, 9\}$ so that (a) The sum of two of the n integers is even. (b) The difference of two of the n integers is 5.
13. How many 3 letters 'words' can be made using the letters a, b, c, d, e, and f if each letter can be used at most once?
14. Find the total number of integers between 1 and 1000 which are neither perfect square nor perfect cubes.
15. Out of 12 employees, a group of four trainees is to be sent for training. In how many ways these can be selected if (a) there are two employees either they both go or both do not go. (b) there are two employees who want to go together and there are two employees who refuse to go together.

16. 10 points on a straight line are joined to 12 points on another straight line. Find the number of points of intersection of the line segments thus formed.
17. Four dice are thrown at a time. In how many ways a total of 16 can be obtained?

18. Find the number of nonnegative integral solution of $x + y + z = 18, x < 7, y < 8, z < 9$.

19. Find the number of ways in which 3 doctors and 15 engineers can sit together in a queue such that between any two doctors at least two engineers can sit.
20. PQR is an equilateral triangle with PQ=1. Show that if at least five points are selected from the interior of PQR, there are at least two points whose distance is less than ½.
21. Let S be a square where each side has length 2 inches. Find the minimum number of points to be chosen from the interior of S such that distance between two of the points will be less than $\sqrt{2}$ inches.

22. Let S be an equilateral triangle where each side has length 3 inches. Find the minimum number of points to be chosen from the interior of S such that distance between two of the interior points will be less than one inch.

23. 6 boys and 6 girls are to be seated in a row. How many ways can they be seated if (i) all boys are to be seated together and all girls are to be seated together, (ii) No two girls should be seated together, (iii) the boys occupy extreme positions.

24. How many three-digit number can be formed from the six digits 2, 3, 5, 6, 7 and 9 if repetitions are not permitted? How many of these are less than 400? How many of these are even?

25. Find the number of divisors of the number 2160. Also, find the sum of these divisors.

26. In a triangle ABC four, five and six points (other than A, B, C) are taken on the sides AB, BC and CA respectively. How many triangles can be constructed using these points as vertices? How many triangles considering the points A, B, C along with the above 15 points as vertices can be constructed?

27. What is the rank of the word "BIPLAB" written down in a dictionary

28. Find the total number of integers lying between 1 and 1000 that are divisible by at least one of 2,3 and 7.

Module-III

1. Show that $\left(q \to (p \wedge {\sim}p)\right) \to \left(r \to (p \wedge {\sim}p)\right) \equiv (r \to q)$.

2. Using the truth table show that $p \to (q \vee r) \equiv (p \to q) \vee (p \to r)$.

3. Find the CNF of the following statement $\sim (q \vee p) \leftrightarrow (q \wedge p)$.

4. Obtain the DNF of $\sim \{\sim (p \leftrightarrow q) \wedge r\}$

5. Prove that the following equivalence $(p \wedge r) \vee (q \wedge r) \vee (\sim p \wedge (\sim q \wedge r)) \equiv r$.

6. Prove that the following equivalence $(q \wedge \sim (p \wedge q)) \vee (p \wedge (\sim p \vee q)) \equiv q$.

7. Construct truth table and determine and determine whether the following proposition is tautology or contradiction $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

8. Find the truth table of $[(p \wedge q) \vee (\sim r)] \leftrightarrow p$.

9. Prove that $\sim(p \vee q) \vee (\sim p \wedge q) \vee p$ is a tautology.

10. Prove the following equivalence: $\sim(p \rightarrow q) \equiv p \wedge \sim q$

11. Show that $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction.

12. Define Tautology. Show that $(p \rightarrow (q \rightarrow p))$ is a tautology.

13. Show that the two statements $(p \rightarrow q) \rightarrow q$ and $p \vee q$ are logically equivalent.

14. Without truth table Prove that $p \leftrightarrow q \equiv (p \vee q) \rightarrow (p \wedge q)$

15. Show that the following is a tautology: $(p \wedge q) \rightarrow (p \vee q)$

Module-IV

1. Let S be the set defined by $S = \{z \in C: |z| = 1\}$, where C is the set of all complex numbers. Show that S forms a commutative group under usual multiplication of complex numbers.
2. Show that all roots of the equation $x^4 = 1$ form an Abelian group under multiplication.
3. Show that the set of rational numbers other than one, $Q' = Q - \{1\}$ forms a group under the binary operation * defined by $a * b = a + b - ab, \forall\, a, b \in Q'$.
4. Show that $G = \{a + b\sqrt{2}, \forall\, a, b \in Q\}$ form a group with respect to multiplication.
5. Prove that the set M of all 2×2 non-singular real matrices form a group under matrix multiplication.
6. Let a be an element of a group $(G, ^\circ)$ of order n. Then $a^m = e$, identity element if and only if n is a divisor of m.
7. If y be an element of a group $(G, ^\circ)$ and $o(y) = 20$, find the order of y^8.
8. Prove that the necessary and sufficient condition of a non-empty subset H of a group $(G, ^\circ)$ to be a sub-group is that $a, b \in H \Rightarrow a^\circ b^{-1} \in H$.
9. Prove that intersection of two subgroups is a subgroup but union of two subgroups may or may not be a subgroup.
10. Define Cyclic group. Prove that every cyclic group is Abelian.
11. Show that the set $G = \{1, 2, 3, 4, 5, 6\}$ forms a cyclic group under the operation multiplication modulo 7. Find all the generators of this group.
12. Show that the group $\{Z_6, +\}$, i. e. the additive group of all integers modulo 6 is cyclic. Find all the generators of the group.
13. Show that the group $\{Z_9, +\}$ is cyclic where $Z_9 = \{[0], [1], [2], [3], [4], [5], [6], [7], [8]\}$. Find all the generators of the group.
14. Find the generators and order of each element of the group $G = \{a, a^2, a^3, \dots, a^8 = e\}$.
15. Show that every proper subgroup of a group of order 6 is cyclic.
16. Using Lagrange theorem, prove that every group of prime order is cyclic.
17. Determine which of the following are even permutation: (a) $f = (1\ 2\ 3)(1\ 2)$ (b) $g = (1\ 2\ 3\ 4\ 5)(1\ 2\ 3)(4\ 5)$.
18. Find the inverse of the permutation: (a) $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ (b) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$.

19. Show that the mapping $f: (Z, .) \to (R, .)$ defined y $f(x) = x^4 \ \forall x \in Z$ is a monomorphism but not isomorphism.
20. Show that the group $\{Z_9, +\}$ is a homomorphic image of the group $\{Z, +\}$.
21. Let $(Q, +)$ be the additive group of rational numbers and$(Q^+, \ .)$ be the multiplicative group of positive rational numbers and $f(x) = 4x$ is a mapping from Q to Q^+. Are these two groups isomorphic? Justify your answer.
22. Show that the set S_3 of all permutations on $\{1,2,3\}$ forms a finite non-abelian group of order 6 with respect to permutation multiplication

23. Let $G = (Z, +)$ and $f: G \to G$ defined by $f(x) = |x|$. Verify whether f is a homomorphism. If so, determine $Ker\ f$.

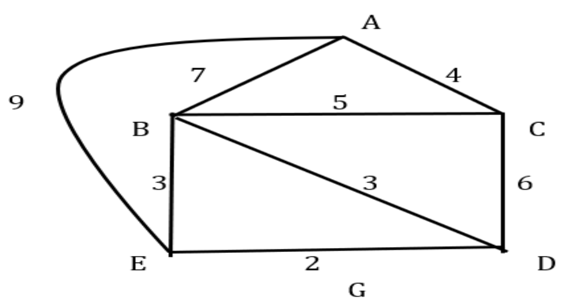
24. Let $G = (Z, +)$ and $f: G \to G$ defined by $f(x) = 4x$. Verify whether f is a homomorphism. If so, determine $Ker\ f$.
25. Define an Integral domain. Show that $(\mathbb{Z}, +, \times)$ is an integral domain.

26. If $f: (C - \{0\}, .) \to (C - \{0\}, .)$; defined by $f(z) = z^4$
    a) Show that f is a homomorphism.
    b) Find the kernel of f.
27. If a ring R with unity,$(xy)^2 = x^2y^2, \forall\ x, y \in R$, then show that R is commutative. Show that a ring R satisfies cancellation laws iff R is without zero divisor.
28. Prove that intersection of two subrings is a subring.
29. Prove that every finite integral domain is a field.

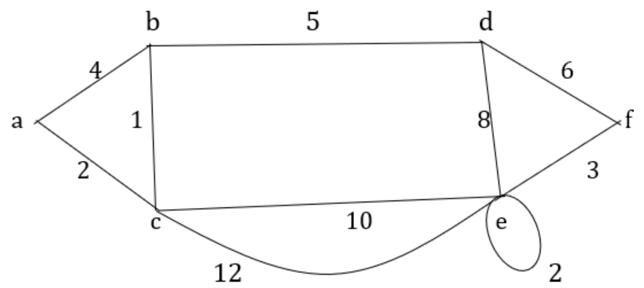
Module-V

1. State and prove Euler's formula for a connected planar graph.
2. Prove that the number of pendent vertices in a binary tree with n vertices is $\frac{(n+1)}{2}$.

3. Prove that a tree with n number of vertices has $(n-1)$ number of edges.
4. State and prove generalized Euler theorem for planar graph.
5. Prove that a connected graph with n vertices and $(n-1)$ edges is a tree.
6. Let G be a simple planar graph having less than 12 vertices. Prove that the graph has a vertex whose degree ≤ 4.
7. Find a minimal spanning tree by the Prims and Kruskal algorithm of the graph G given below:



a.



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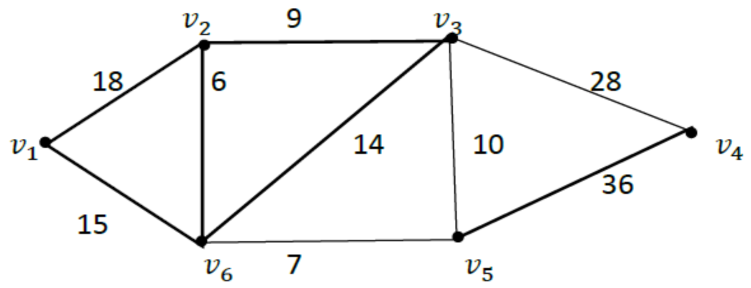
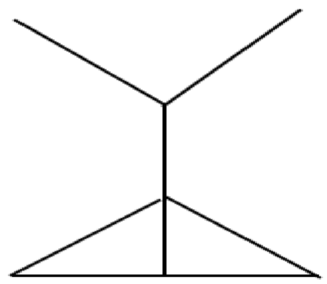


8. Draw the dual of the following graph

(i)



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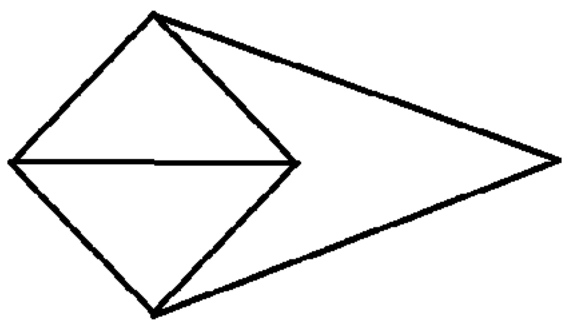
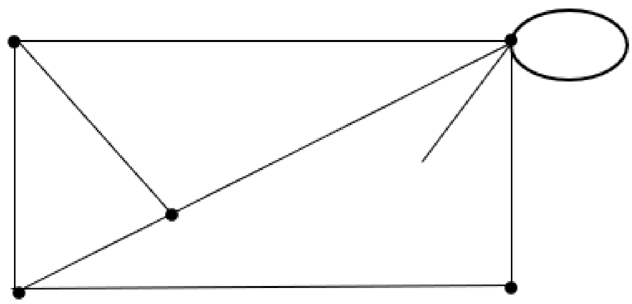
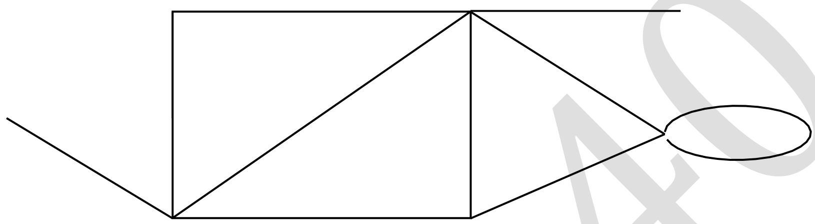


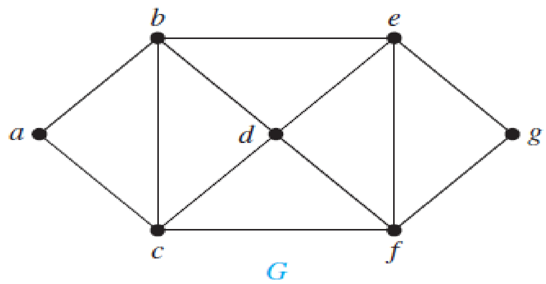
(iii)



(iv)

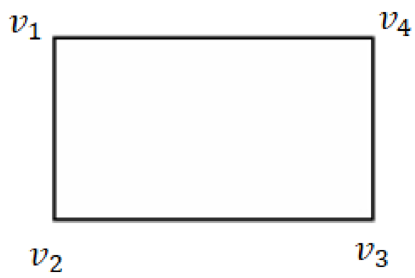


9. Find three different independent set of vertices with different size and also find the chromatic number of the graph given below

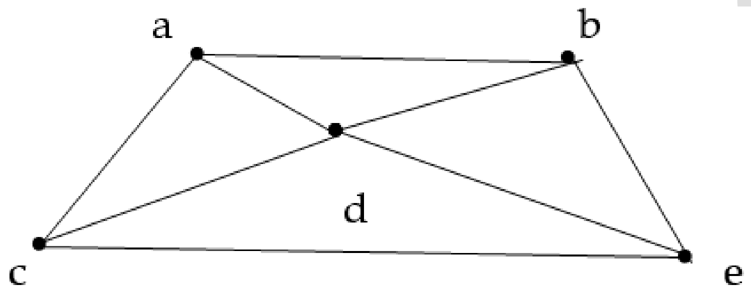


10. Find the chromatic polynomial of the given graph, hence find its chromatic number.

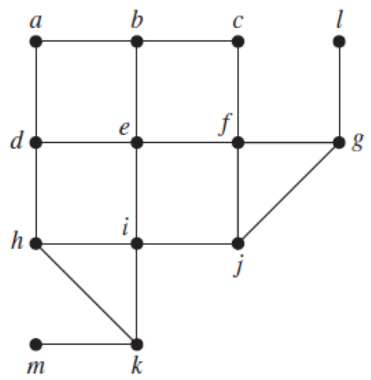
(i)



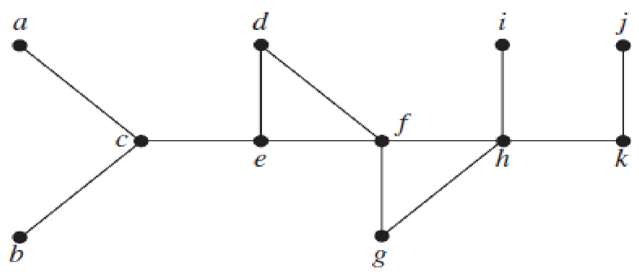
(ii)



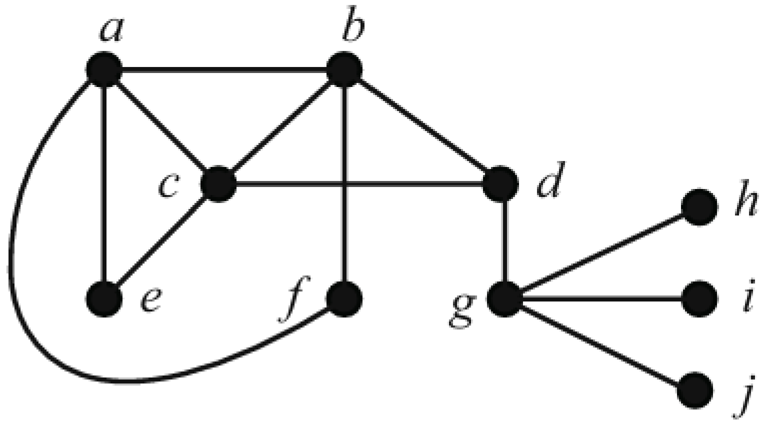
11. Find a spanning tree starting from **d** for the following graph using **breadth-first search** and **depth-first search**
    (i)



(ii)



12. Find two different matchings of different size of the following graph. Also find maximum matching and maximal matching of the following graph.



13. Applicants a_1, a_2, a_3, a_4 apply for five posts p_1, p_2, p_3, p_4 and p_5. The applications are done as follows: $a_1 \rightarrow \{p_1, p_2\}$, $a_2 \rightarrow \{p_1, p_3, p_5\}$, $a_3 \rightarrow \{p_1, p_2, p_3, p_5\}$ and $a_4 \rightarrow \{p_3, p_4\}$. (i) Draw a bipartite graph of the relationship between applicants and posts (ii) Is there any perfect matching of the set of applicants into the set of post? If yes use Hall's marriage theorem to show that every applicant can be offered a single post.

14. Suppose there are 3 boys a, b, c knows four girls x, y, z, w as follows:

Boy	Girls
A	w, y, z
B	x, z
c	x, y

Draw a bipartite graph corresponding to the table, (ii) Find five different matchings, (iii) Show that the Hall's marriage condition is satisfied for the above matching.